

LESSON 319 (1.0 AATD)

Lesson Objective

During this lesson, the student will review holding patterns. The student will also be introduced to VOR, front course localizer, and ILS approaches.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Simulator)

To save time, consider skipping the instrument cockpit check and starting from the air. Use the sim's "Aircraft Attitude" menu to set their altitude, heading, and airspeed. For this first sim, set the wind to be little to none.

GPS Holding Procedures

- Non-standard hold:
 - Have the student proceed to a GPS fix and hold until "approach is ready to give practice approaches" (for example, Hold SW of VPBOV on a 230 bearing, left turns).
 - Ensure they follow the correct procedures by using the 6 Ts and applying wind and time corrections as necessary.
 - After the entry, have the student depart on the outbound leg of the hold to save time.

Instrument approach procedures

- **VOR approach - followed by a missed approach (VOR hold):**
 - Tell the student to expect a VOR approach starting from a VOR (for example, the VOR 09R full approach at KMLB). Give them some space then tell them to proceed direct to the VOR, then fly outbound.
 - Once established on this track, consider engaging the autopilot or pausing the sim to allow the student to run the WIRE checklist.
 - Ensure the student completes all the steps of the WIRE checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS)
 - Load the procedure into the GPS. Load each procedure from an IAF transition (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
 - Explain the difference between LOAD vs ACTIVATE. Loading the approach preserves waypoints in the flight plan to allow for an activation once ready to commence the approach. Activating the

Instrument Rating Course

approach deletes all other waypoints and goes direct to whichever transition was selected.

■ Radios:

- Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
- Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
- Set the final approach course if required.

■ Enroute brief

- Prior to reaching the VOR, point out the G430's prompts to turn early onto the outbound radial so as not to overshoot. Turn to the outbound heading and twist to the outbound radial.
- Emphasize that since an approach clearance has not yet been given, a descent or procedure turn is not authorized. Shortly after, clear them for the approach.
- Ensure the student utilizes proper descent planning when deciding when to execute their procedure turn.
- During the procedure turn, ensure the student sets airspeed to 90 knots, sets flaps to 10°, sets the MFD to the engine page, and sets the inbound course.
- Explain that since there is no glideslope, a proper descent rate can be maintained by setting approximately 1,800RPM (1,500RPM in the simulator) and pitching for 90 knots.
- Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
- Ensure the student levels off at and holds the MDA until the MAP. Remind the student of the 91.175(c) criteria required to land.
- Ensure the student completes the steps for a missed approach correctly:
 - Set full power and pitch up for V_y
 - Upon reaching a positive rate of climb, retract the flaps
 - Turn to the missed approach heading/radial
 - Notify ATC
- Have the student do one lap in the published hold, then provide a radar vector for the next approach.

● Localizer approach:

- Assign the student a heading and tell them to expect radar vectors for a localizer approach (for example, the LOC 09R at KMLB).
- Once established on this vector, consider engaging the autopilot or pausing the sim to allow the student to run the WIRE checklist.
- Ensure the student completes all the steps of the WIRE checklist:
 - Weather
 - Instruments:

Instrument Rating Course

- Set both altimeters.
- Set the correct nav source on the PFD (VLOC/GPS)
- Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
- Radios:
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course.
- Enroute brief
 - Fail the glideslope using the Instruments menu.
 - Give the student a vector to final along with an approach clearance. This is a good time to ensure airspeed is 90 knots, extend the flaps to 10°, and set the MFD to the engine page.
 - Explain that since there is no glideslope, a proper descent rate can be maintained by setting approximately 1,800RPM (1,500RPM in the simulator) and pitching for 90 knots.
 - Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
 - Ensure the student levels off at and holds the MDA. Remind the student of the 91.175(c) criteria required to land.
 - Ensure the student executes the steps of a missed approach correctly:
 - Set full power and pitch up for Vy
 - Upon reaching a positive rate of climb, retract the flaps
 - Turn to the missed approach heading/radial
 - Notify ATC
- ILS approach:
 - Assign the student a heading and tell them to expect radar vectors for an ILS approach (for example, ILS 09R at KMLB).
 - Once established on this vector, consider engaging the autopilot or pausing the sim to allow the student to run the WIRE checklist.
 - Ensure the student completes all the steps of the WIRE checklist:
 - Weather
 - Instruments:
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS).
 - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.

- Set the final approach course if required.
- Radios:
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
- Enroute brief
 - Set the weather to allow for a landing.
 - Give the student a vector to final along with an approach clearance. This is a good time to ensure airspeed is 90 knots, extend the flaps to 10°, and set the MFD to the engine page.
 - Explain that setting around 1,800RPM in the aircraft (1,500RPM in the sim) will provide a roughly -500fpm descent rate at 90 knots.
 - Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
 - Remind the student of the 91.175(c) required to land. Once these criteria are met, extend the flaps and land.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 320
- Assign Homework:
 - Review procedures for DME arcs.
 - Review procedures for localizer back course and RNAV approaches.

Completion Standards

At the completion of this lesson, the student should be able to do the following with minimal instructor assistance:

1. Explain and use the information displayed on the approach charts.
2. Execute several initial and intermediate approach segments to arrive at the final approach fix.
3. Complete the final approach segment and descent to the missed approach point.
4. Demonstrate the missed approach procedure, as published on the appropriate chart.